

Business Requirements Document (BRD)

Cart Ninja — Shopify Cart Enhancement App

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1. Executive Summary

Cart Ninja is a Shopify application designed to increase merchant revenue by enhancing the default Shopify cart experience. It replaces the native cart with a feature-rich, customizable slide-out cart drawer that integrates rewards gamification, AI-powered upselling, coupon management, and Frequently Bought Together (FBT) recommendations. The app provides merchants with a unified dashboard to configure all features and track performance through a real-time analytics suite.

2. Business Objectives

#	Objective
1	Increase Average Order Value (AOV) for Shopify merchants
2	Improve cart-to-checkout conversion rate
3	Provide merchants a no-code solution to deploy cart enhancements
4	Enable data-driven decisions through actionable cart analytics
5	Maximize coupon redemption and upsell click-through rates

3. Scope

3.1 In Scope

- Shopify Admin dashboard application (merchant-facing configuration)
- Theme App Extension injected into the storefront (customer-facing cart UI)
- Rewards progress bar with multi-tier milestone gamification
- AI and manual upsell product recommendation engine
- Coupon slider widget with Shopify discount sync
- Frequently Bought Together (FBT) product widget
- Real-time analytics dashboard with date-range filtering
- Billing and subscription management
- GDPR-compliant webhook handlers

3.2 Out of Scope

- Custom storefront (Hydrogen/Headless) support
- Native mobile app (iOS / Android)
- Email or SMS marketing integrations
- Inventory management
- Multi-store management panel

4. Stakeholders

Role	Responsibility
Product Owner	Define feature requirements and acceptance criteria
Developer	Build, test, and deploy the application
Shopify Merchant	End-user configuring and operating the app
Shopify Customer	End-consumer interacting with the cart drawer on storefront
Analytics Backend (int.thecartninja.com)	Aggregates and serves event data

5. User Personas

5.1 Merchant (Primary User)

- Shopify store owner or manager
- Non-technical; expects a no-code setup via a guided dashboard
- Goal: Increase revenue without developer involvement
- Needs: Easy configuration, clear analytics, reliable coupon/upsell management

5.2 Shopify Customer (End Consumer)

- Visitor browsing and shopping on a merchant's storefront
 - Interacts with cart drawer, reward milestones, coupon codes, and upsell offers
 - Goal: Complete purchase; optionally discover relevant products or redeem discounts
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6. Functional Requirements

6.1 Cart Drawer

ID	Requirement
CD-01	Replace the default Shopify cart with a customizable slide-out drawer
CD-02	Display cart line items with quantity controls and remove buttons
CD-03	Show real-time cart subtotal in the store's currency
CD-04	Provide a customizable Checkout button with configurable label and color
CD-05	Support empty cart state with configurable messaging
CD-06	Render responsively on desktop and mobile
CD-07	Allow merchants to configure border radius, background color, and text colors
CD-08	Inject into storefront via Shopify Theme App Extension without theme code edits

6.2 Rewards Progress Bar

ID	Requirement
RP-01	Display a visual progress bar showing proximity to the next reward milestone
RP-02	Support up to 5 configurable milestone tiers
RP-03	Support two progress modes: cart value (amount) and cart quantity (items)
RP-04	Allow per-tier configuration: icon, reward type, title, and messaging
RP-05	Provide icon presets: gift, shipping, discount, star, trophy, heart, diamond, lock; and support custom SVG icons
RP-06	Show "almost there" message as customer approaches a milestone
RP-07	Trigger confetti animation (toggleable) upon milestone completion
RP-08	Support gradient fill styling on the progress bar
RP-09	Allow merchants to set a custom completion message per tier

6.3 Upsell Engine

ID	Requirement
US-01	Provide AI-powered product recommendation mode
US-02	Provide a manual rules engine with three rule types
US-03	Rule Type 1 — Global: Display selected products to all customers
US-04	Rule Type 2 — Trigger-based: Display products only if specific products or collections are in the cart
US-05	Rule Type 3 — Cart Value: Display products only if cart subtotal exceeds a configured threshold
US-06	Allow carousel or grid layout selection
US-07	Allow configuration of maximum number of products to display
US-08	Allow toggling product reviews/ratings display
US-09	Allow configuring custom Add-to-Cart button label
US-10	Allow showing upsell section even when cart is empty (optional)

6.4 Frequently Bought Together (FBT)

ID	Requirement
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FBT-01	Display complementary product suggestions based on cart contents
FBT-02	Show product image, title, price, and quick-add button per suggestion
FBT-03	Configurable as a standalone storefront widget
FBT-04	Log FBT interaction events to the analytics backend

6.5 Coupon Slider

ID	Requirement
CPN-01	Sync available coupons from Shopify Admin API (Basic %, BXGY, Free Shipping)
CPN-02	Display coupons in three visual style presets
CPN-03	Support per-coupon style overrides (color, icon, background, text)
CPN-04	Allow positioning the coupon section at top or bottom of drawer
CPN-05	Support grid and carousel layout options
CPN-06	Support horizontal and vertical alignment
CPN-07	Enable one-click coupon application to the cart (exclusive — one active at a time)
CPN-08	Allow configuring section title with font size, color, and alignment
CPN-09	Track coupon click events to the analytics backend

6.6 Analytics Dashboard

ID	Requirement
AN-01	Display KPI cards: Checkout Clicks, Coupon Clicks, Upsell Clicks, Upsell Revenue, Total Cart Revenue, Total Coupons Applied
AN-02	Provide a date range picker with presets: Today, Yesterday, Last 7 / 30 / 90 Days, Last 12 Months
AN-03	Support Fixed and Rolling date range modes
AN-04	Render an area chart for revenue trends over time
AN-05	Render a bar chart for performance breakdown by metric
AN-06	Render a composed chart (bars + line) for click-through rate
AN-07	Render a pie chart for cart journey breakdown (abandoned vs. completed)

AN-08	Format all revenue values using the shop's currency and locale
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6.7 Coupon / Discount Management

ID	Requirement
DM-01	List all Shopify discounts with status (Active, Scheduled, Expired), type, and source
DM-02	Filter discounts by status, type, and source
DM-03	Sort discounts by creation date or usage count
DM-04	Allow deletion of app-created coupons
DM-05	Display usage count versus usage limit per coupon
DM-06	Provide a form to create new discount coupons via Shopify GraphQL mutations

6.8 Billing & Subscription

ID	Requirement
BL-01	Display a subscription plan selection screen
BL-02	Show billing history with charge dates and amounts
BL-03	Support one-time charge triggers via Shopify Billing API
BL-04	Handle subscription update webhooks from Shopify

6.9 Onboarding / Setup

ID	Requirement
SU-01	Provide a step-by-step installation guide
SU-02	Include deep links to the merchant's Shopify theme editor
SU-03	Track onboarding completion steps

6.10 Webhooks & Compliance

ID	Requirement
WH-01	Handle <code>app/uninstalled</code> webhook to clean up merchant data
WH-02	Handle <code>app/subscriptions/update</code> webhook
WH-03	Handle <code>customers/redact</code> webhook for GDPR compliance
WH-04	Handle <code>shop/redact</code> webhook for GDPR compliance
WH-05	Handle <code>app/scopes/update</code> webhook for OAuth scope changes

7. Non-Functional Requirements

ID	Category	Requirement
NF-01	Performance	Cart drawer should render and become interactive within 300ms of page load
NF-02	Availability	Admin dashboard and API endpoints must maintain 99.9% uptime
NF-03	Security	All admin routes must be authenticated via Shopify OAuth; API routes must validate shop session
NF-04	Scalability	Architecture must support horizontal scaling; database should support PostgreSQL in production
NF-05	Compatibility	Cart drawer must support the latest two major versions of Chrome, Firefox, Safari, and Edge
NF-06	GDPR	All customer data must be erasable on merchant request via GDPR webhooks
NF-07	Accessibility	Admin dashboard UI must follow Shopify Polaris accessibility guidelines
NF-08	Responsiveness	Storefront cart drawer must function correctly on viewport widths from 320px to 2560px

8. Technical Architecture

8.1 Tech Stack

Layer	Technology
Frontend Framework	React 18.3 with React Router 7.x
Admin UI Library	Shopify Polaris v13.9.5
Database ORM	Prisma 6.x
Database (Dev)	SQLite
Database (Prod)	PostgreSQL / MySQL
Charts	Recharts 3.7.0
Shopify Integration	@shopify/shopify-app-react-router, App Bridge React
Theme Extension	Shopify Theme App Extension (Liquid + Vanilla JS/CSS)
Analytics Backend	Remote PHP service at https://int.thecartninja.com

8.2 Database Models

Model	Purpose
Session	Stores Shopify OAuth sessions and admin credentials
UpsellRule	Stores manual upsell configurations per shop (GLOBAL, TRIGGERED, CART_CONDITIONS)
WidgetSettings	JSON-based storage for all widget configurations (coupons, FBT, progress bar, upsell) per shop

8.3 Key Integration Flows

1. **Merchant Configuration** → Admin Dashboard → Prisma DB + Remote PHP Backend
2. **Theme Activation** → Merchant enables App Embed in Shopify Theme Editor
3. **Storefront Load** → Liquid block injects JS/CSS → JS fetches config via `/api/cartdrawer-config`
4. **Customer Interaction** → Cart events → Analytics tracking to `https://int.thecartninja.com`
5. **Analytics View** → Dashboard fetches aggregated data via `/api/analytics`

9. User Stories

ID	As a...	I want to...	So that...
S-01	Merchant	Configure a slide-out cart drawer with my brand colors	My cart matches my store's visual identity
S-02	Merchant	Set up milestone rewards for customers	Customers are motivated to add more items
S-03	Merchant	Create upsell rules to show relevant products	I can increase AOV without manual merchandising
S-04	Merchant	Sync my Shopify coupons into the cart drawer	Customers can easily discover and apply discounts
S-05	Merchant	View analytics on coupon usage and upsell clicks	I can measure the ROI of my cart optimizations
S-06	Merchant	Create new discount coupons directly from the app	I don't have to leave the app to manage promotions
S-07	Customer	See a rewards progress bar in the cart	I know how much more I need to spend for a reward
S-08	Customer	Apply a coupon code with one click	Checkout is faster and more convenient
S-09	Customer	See recommended products in the cart	I can discover relevant items before checking out
S-10	Customer	View FBT suggestions for products in my cart	I can complete my purchase more easily

10. Assumptions & Constraints

Assumptions

- Merchants are using online Shopify stores (not POS-only)
- Merchants have at least one active Shopify theme
- The remote analytics backend (<https://int.thecartninja.com>) is maintained separately
- Shopify App Store listing and review are handled outside this document's scope

Constraints

- App must comply with Shopify's Partner Program policies
- All storefront-side code must be injected via Theme App Extension (no direct theme file edits)
- GDPR webhook handlers must respond within Shopify's required timeframes
- App billing must use Shopify's native Billing API

11. Acceptance Criteria

Feature	Acceptance Criteria
Cart Drawer	Cart opens on cart icon click; items display with correct prices; checkout button routes to Shopify checkout
Progress Bar	Bar fills proportionally to cart value/quantity; milestone icon and message update at each tier
Upsell	Products matching configured rules appear in carousel/grid; clicking Add adds item to cart
Coupons	Synced coupons appear in drawer; clicking a coupon applies it and shows confirmation
FBT	Relevant products appear below cart items; quick-add button works without page reload
Analytics	Dashboard loads within 3 seconds; all KPIs match data from analytics backend for selected date range
Billing	Merchant can view plan options and upgrade; billing history loads accurately

12. Risks & Mitigations

Risk	Probability	Impact	Mitigation
Shopify API breaking changes	Medium	High	Pin API version; monitor Shopify changelog
Analytics backend downtime	Low	Medium	Graceful degradation; dashboard shows cached/empty state
Theme conflict with cart drawer injection	Medium	High	Test on top 10 Shopify themes; provide conflict resolution guide
App Store rejection	Low	High	Follow Shopify review guidelines; test on dev store before submission
Data loss on uninstall	Low	High	Webhooks clean up data; merchants warned before uninstall

13. Glossary

Term	Definition
AOV	Average Order Value — average spend per transaction
FBT	Frequently Bought Together — product recommendation strategy
Cart Drawer	A slide-out panel replacing the default Shopify cart page
Theme App Extension	Shopify mechanism for injecting app code into storefronts without editing theme files
Upsell	Suggesting a higher-value or complementary product to increase order value
Milestone	A reward threshold in the progress bar system
KPI	Key Performance Indicator
GDPR	General Data Protection Regulation (EU data privacy law)

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